Google Workspace + Gemini Enterprise Target Architecture Blueprint (Enterprise/150k+)

Implementation Solution Guide

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Executive Summary

The problem

In a 150,000-user shift from Microsoft 365 to Google Workspace and Gemini Enterprise, whoever owns the **target architecture** owns the outcome. When no one owns the end state, each partner and each migration wave invents its own design: identity fragments across shadow directories, sharing and DLP drift, coexistence bridges never sunset, and AI enablement outruns governance. The result is architectural rework wave after wave — the most expensive failure mode in an enterprise transformation.

The approach

This blueprint defines and keeps authoritative the **end-state Google Workspace + Gemini + identity architecture** — and the principles that govern every workstream. It is an architecture-authority product: **principles, reference patterns, architecture decision records (ADRs), and non-functional requirements**, not a click-by-click runbook.

The architecture is organized into five domains, designed in dependency order:

1. **Identity integration** — one authoritative source, one provisioning path, federated SSO

(the spine every other domain resolves access through).

1. **Collaboration** — Google Workspace as the single email/calendar/Chat/Meet plane, with bounded,

sunsetting coexistence.

1. **Drive / content replacement** — shared drives as the SharePoint/OneDrive successor, migrated by

content class with permissions remodeled, not copied.

1. **Data governance & retention** — Vault + Workspace DLP + the AI control center as a shared

governance plane, with a bounded Purview→Vault hold transition.

1. **AI enablement** — Gemini Enterprise and governed agents, enabled by scope and **only where the

data-class governance already exists**.

Nine enterprise principles (one owned end-state, identity-first, govern-by-scope, least standing access, bounded coexistence, governance-before-AI, avoid needless lock-in, NFRs first-class, and the Assess → Configure → Pilot → Validate → Expand lifecycle) constrain every design and are enforced by an **Architecture Review Board** through pass/fail conformance gates.

What's included

The architecture-definition method across all five domains; a reusable ADR/NFR/standards framework; five source-side PowerShell scripts (current-state discovery, conformance validation, coverage/gap report, safe remediation guidance, and scope targeting — read-only where they touch live systems); and a **Target-Architecture document template outline** you populate to produce the authoritative artifact the program governs against.

> ***Vendor-accuracy note.*** *Google Admin console and Microsoft/Entra patterns are drawn from current public vendor documentation;* ***Gemini Enterprise capabilities that are preview or rolling out are***

> ***flagged "verify in your tenant"*** *and carried in a preview-capability risk register — never assumed GA.*

The bottom line

An approved, stable enterprise target architecture — every workstream aligned to one end-state design, every decision recorded, every wave gated by the ARB — so migration waves execute **without architectural rework**. Define it, own it, align every workstream to it. Self-guided and validated, at a fraction of the cost of open-ended architecture consulting.

Solution Overview — Target Architecture Blueprint

What this blueprint is (and is not)

This is an **architecture-authority** product for a large-scale (150,000-user) transformation from Microsoft 365 to **Google Workspace + Gemini Enterprise**. It defines the **end-state target architecture** and the **architectural principles, reference patterns, decision records (ADRs), and non-functional requirements (NFRs)** that every migration workstream is measured against. It is the document the chief architect owns so the transformation stays scalable, supportable, and under enterprise control.

It is **not** a click-by-click configuration runbook. Individual workstreams (mail cutover, Drive migration, SSO federation, DLP, Vault) each have their own build procedures; this blueprint governs **how those builds must be designed** so they compose into one coherent, defensible end state instead of fragmenting across partners and waves.

> ***Authority statement.*** *When a workstream design conflicts with a principle or ADR in this blueprint, the blueprint wins unless the Architecture Review Board (ARB) records a superseding decision. That is what "owning the architecture" means in practice.*

The end-state architecture at a glance

The target estate is organized into five **architecture domains**, each with an owning principle set, reference patterns, and ADRs (detailed in `03-implementation.md`):

Domain	End-state anchor	Source-side coexistence
Collaboration (email, calendar, Chat, Meet, Docs)	Google Workspace as the single collaboration plane	M365 mailboxes decommissioned wave-by-wave; dual-delivery only during coexistence
Content / Drive replacement	Google Drive + shared drives as the SharePoint/OneDrive successor	SharePoint/OneDrive read-only then retired per content class
Identity integration	One authoritative identity source federated to Google via SSO + automated provisioning	Entra ID / on-prem AD retained as source of truth or transitional IdP
Data governance & retention	Google Vault + Workspace DLP + AI control center as the governance plane	Microsoft Purview retained for legacy holds until legal sunset

Domain	End-state anchor	Source-side coexistence
AI enablement	Gemini Enterprise + agentic workflows governed by the AI control center	N/A (net-new capability)

These five domains sit on a **shared foundation**: the identity source of truth, the organizational unit (OU) and group model, DNS/mail-flow control, and change control. Every domain design must trace back to that foundation — this is the **GWARCH-Foundation** principle set (see below).

How the workstreams fit together

The transformation runs many parallel workstreams (mail, calendar, Drive, Chat, identity, governance, AI, endpoint/Chrome). Left uncoordinated they drift. The blueprint binds them with three integration contracts:

1. **Identity contract.** Every domain consumes users and groups from **one** provisioning pipeline

(see the *Identity integration* domain). No workstream provisions its own shadow identities. This is the **GWARCH-ID-1** principle: *one authoritative identity source, one provisioning path.*

1. **OU/group targeting contract.** Every policy (DLP, sharing, Gemini access, retention) is applied

to the **same** named OU/group taxonomy — the **GWARCH-Scope** model — so scope is consistent and auditable across domains. `GWARCH-Set-ScopeTargeting.ps1` operationalizes this on the source side.

1. **Governance contract.** Every domain exposes its policy surface to the governance plane (Vault,

DLP, AI control center) rather than governing itself in isolation. This is **GWARCH-GOV-1**: *governance is a shared plane, not a per-app setting.*

Enterprise architecture principles (the constraints on every design)

These principles constrain all five domains. Each is named so downstream designs and ADRs can cite it.

• **GWARCH-P1 — One end-state, owned centrally.** There is a single approved target architecture;

partner/wave designs conform to it or raise an ARB exception. No workstream defines its own end state.

• **GWARCH-P2 — Identity is the spine.** Collaboration, content, governance, and AI all resolve

access through one authoritative identity and one provisioning pipeline (GWARCH-ID-1). Identity decisions are made first and constrain everything else.

• **GWARCH-P3 — Govern by scope, not by exception.** Policy is applied to the named OU/group

taxonomy (GWARCH-Scope), never to ad-hoc individuals. New capabilities inherit governance by landing in the right scope.

• **GWARCH-P4 — Least standing access; sharing is a decision, not a default.** External sharing,

broad admin roles, and AI data access are closed by default and opened per scope with justification.

• **GWARCH-P5 — Coexistence is bounded and temporary.** Every M365↔Google coexistence bridge (mail

routing, directory, holds) has a defined sunset condition. Nothing "temporary" becomes permanent without an ARB decision.

- **GWARCH-P6 — Data governance precedes AI enablement.** Gemini/agent access to Workspace data is

only enabled where DLP, sharing, and retention controls for that data class are already in place (GWARCH-GOV-1). AI does not outrun governance.

- **GWARCH-P7 — Avoid one-way lock-in where reversible costs little.** Prefer open, exportable data

formats and portable identity patterns so a future change of tooling touches integration layers, not the whole estate.

- **GWARCH-P8 — Non-functional requirements are first-class.** Availability, scale (150k users),

supportability, security, and compliance NFRs are stated per domain and validated at conformance gates, not assumed.

- **GWARCH-P9 — Assess → Configure → Pilot → Validate → Expand.** Every domain change follows this

model with explicit pass/fail gates and a documented rollback for production-affecting changes.

Deployment / decision model

The blueprint itself is delivered and kept authoritative through the same lifecycle the workstreams use:

Assess current-state (source M365/Entra discovery via `GWARCH-Discovery.ps1`) → **Configure** the target architecture (principles, domain patterns, ADRs, NFRs) → **Pilot** the architecture against one wave/OU and validate conformance (`GWARCH-Validation.ps1`) → **Validate** coverage and close gaps (`GWARCH-Report.ps1`, `GWARCH-Remediation.ps1`) → **Expand** the approved architecture to all waves under ARB control.

Who uses this

Chief/enterprise architect and the ARB (own and enforce it); workstream leads (design to it); security, legal/eDiscovery, and data-governance stakeholders (validate their controls are expressed in it); program leadership (measure wave readiness against it). Audience is **senior**: the document assumes fluency in identity, collaboration, and governance concepts and is decision-oriented, not tutorial.

What you get

The `source/` docs (overview, requirements, the architecture-definition method, conformance testing, troubleshooting, best practices, deliverables, executive summary), five source-side PowerShell scripts (discovery, validation, coverage/gap report, remediation guidance, scope targeting), and a **Target-Architecture document template outline** (`08-deliverables-summary.md`) you populate to produce the authoritative artifact your program governs against.

Requirements

These are the prerequisites for **defining and owning** the target architecture — not for executing any single workstream. The category-wide prerequisites in `global-requirements-google.md` (Google tenant, source M365/Entra access, DNS control, identity source of truth, pilot scope, governance

stakeholders) apply in full and are assumed here. This section adds the requirements specific to running the architecture-authority function. Each follows **Description / Why / Verify / Implement if missing**.

1. Named target and source tenants, with edition clarity

• **Description:** A provisioned **Google Workspace** tenant (target) at the edition the end state requires — Enterprise Plus (or the current top-tier equivalent) for Vault, context-aware access, and enterprise DLP; the **Gemini/AI add-on** where AI enablement is in scope — and administrative read access to the **source Microsoft 365 / Entra ID** estate.

• **Why:** The architecture must be designed against the *actual* license entitlements and tenant configuration on both sides, not against assumptions. Edition determines which governance and AI controls are even available to design around.

• **Verify:** In **admin.google.com** → **Billing/Subscriptions**, confirm edition and seat counts; confirm the Gemini/AI subscription state (note: some Gemini Enterprise capabilities are preview/rolling-out — **verify in your tenant**). On the source side, confirm read access to the M365 admin center and Entra admin center.

• **Implement if missing:** Provision/true-up licenses via Google or a reseller; obtain least-privilege read roles on the source (Global Reader on Entra is sufficient for discovery).

2. A single, agreed identity source of truth

• **Description:** One authoritative directory (on-prem AD, Entra ID, or an HR/IGA system) formally designated as the source of truth for the target estate, and the intended provisioning path into Google (Entra provisioning/SCIM, Google Cloud Directory Sync, or Admin SDK).

• **Why:** Principle **GWARCH-P2 / GWARCH-ID-1** makes identity the spine of the architecture. The entire design is undefined until "who is authoritative, and how do objects flow into Google" is settled. Competing sources of truth are the single biggest cause of enterprise-migration rework.

• **Verify:** A written statement naming the authoritative directory and the provisioning mechanism, signed off by the identity owner. Confirm no workstream plans to provision users independently.

• **Implement if missing:** Convene the identity owner and decide the source of truth **before** any domain design proceeds. Record the decision as ADR **GWARCH-ADR-ID-01**.

3. OU and group taxonomy (the GWARCH-Scope model)

• **Description:** A defined organizational-unit and group taxonomy in Google that all policy targets (collaboration settings, sharing, DLP, retention, Gemini access) will bind to — including at least a **pilot** scope, per-wave scopes, and any regulated/high-sensitivity scope.

• **Why:** Principle **GWARCH-P3** requires governing by scope, not by exception. A stable, named taxonomy is what makes policy consistent and auditable across all five domains, and lets `GWARCH-Set-ScopeTargeting.ps1` map source groups to target scopes deterministically.

• **Verify:** The OU tree and control groups exist (or are designed) with documented naming; a pilot OU/group is named.

• **Implement if missing:** Design the taxonomy before configuring any policy; align source-side groups so scope maps cleanly. Record as ADR **GWARCH-ADR-SCOPE-01**.

4. Governance, security, legal & compliance stakeholders engaged

• **Description:** Named owners for **security (CISO/DLP), legal & eDiscovery (Vault/holds), data governance (retention/classification), and any regulatory regime** in scope, engaged as architecture stakeholders — not merely consulted at the end.

• **Why:** Principle **GWARCH-P6** makes governance a precondition for AI enablement and a first-class domain. Retention, hold, DLP, and classification requirements are *inputs* to the architecture; the design cannot be finalized without their sign-off.

• **Verify:** A stakeholder RACI naming each owner and the policies (retention schedules, DLP rules, hold obligations, regulatory controls) the architecture must satisfy.

• **Implement if missing:** Identify owners and collect their policy requirements before the data governance/retention and AI domains are designed.

5. An Architecture Review Board (ARB) and decision-record process

• **Description:** A standing **Architecture Review Board** with a defined cadence and a lightweight **ADR** process for recording decisions, exceptions, and supersessions against the blueprint.

• **Why:** The blueprint is only authoritative if there is a body that enforces conformance and a record that captures why deviations were allowed. Without the ARB, "owning the architecture" has no teeth (principle **GWARCH-P1**).

• **Verify:** ARB charter, membership, cadence, and an ADR log location exist.

• **Implement if missing:** Charter the ARB and stand up the ADR log (the templates in `08-deliverables-summary.md` seed both) before Pilot.

6. Bounded coexistence design inputs (DNS, mail flow, directory bridge)

• **Description:** Control over DNS for routed domains and a documented intended **coexistence** model for the transition period — mail routing/dual-delivery, directory bridging, and legacy holds — each with a **sunset condition**.

• **Why:** Principle **GWARCH-P5** requires coexistence to be bounded and temporary. These bridges are high-impact (mail flow, identity) and must be designed, change-controlled, and given an explicit end date, not left to accrete.

• **Verify:** Access to the authoritative DNS zones is confirmed; a coexistence design with sunset

criteria per bridge is drafted.

• **Implement if missing:** Confirm DNS control and draft the coexistence model with sunset conditions under change control before cutover waves begin.

7. Current-state discovery tooling and access

• **Description:** A workstation able to run the **Microsoft Graph PowerShell SDK** against the source M365/Entra tenant (read scopes), plus super-admin/read access to admin.google.com for target-side confirmation.

• **Why:** The Assess phase depends on repeatable, read-only current-state discovery (`GWARCH-Discovery.ps1`) to ground the architecture in reality rather than assumption (principle **GWARCH-P8**).

• **Verify:** `Connect-MgGraph` succeeds with the read scopes listed in each script header; the scripts run and emit their reports.

• **Implement if missing:** Install the Graph SDK (`Install-Module Microsoft.Graph`), consent the read scopes, and confirm target-side console access.

Deployment/decision model reminder: every requirement above is validated in **Assess**; the architecture is authored in **Configure**; the requirements' sufficiency is proven in **Pilot** and **Validate** before **Expand**.

Global Requirements — Google Workspace & Gemini Solutions

These requirements apply to **every** solution in the **Google** category. They are the common foundation each Google Workspace / Gemini solution assumes is already in place (or stands up as part of its own Requirements section). Verify each before starting a solution's implementation steps.

> ***Vendor boundary (applies to every Google solution).*** *The **Google Admin console** (admin.google.com), **Google Cloud / Cloud Identity**, and the **Microsoft / Entra** sides are authored at exact, current click-by-click detail from the vendors' public documentation. **Gemini Enterprise** capabilities that are in preview or rolling out are **cited or flagged "verify in your tenant"** — never invented. **Third-party migration tools** (e.g. for mail/Drive/Chat data movement) and **third-party enterprise apps** (ServiceNow, SAP, Databricks, etc.) are integrated at the pattern/architectural level; their internal console steps are deferred to the vendor's own docs.*

1. Google Workspace / Cloud Identity tenant

• **Description:** A provisioned Google Workspace (or Cloud Identity) tenant with a **super-admin** account for configuration, and the appropriate Workspace edition/licensing for the capability in scope (e.g. Enterprise Plus for Vault/CAA/DLP, Gemini/AI add-on where required).

• **Why required:** All configuration, policy, and API work is performed against this tenant.

• **How to verify:** Sign in to **admin.google.com** as a super-admin; confirm edition and license counts under **Billing / Subscriptions**.

• **How to implement if missing:** Provision the tenant and licenses through Google or an authorized reseller before starting.

2. Source Microsoft 365 / Entra environment (for migration & coexistence)

• **Description:** For migration/coexistence/identity solutions, administrative access to the source **Microsoft 365** tenant and **Microsoft Entra ID** (Global Administrator or the least-privilege roles the solution names).

• **Why required:** Migration, directory sync, SSO federation, and coexistence read from and change the Microsoft side.

• **How to verify:** Confirm access to the M365 admin center and the Entra admin center with the required role.

• **How to implement if missing:** Obtain the role from the M365/Entra tenant owner.

3. DNS control for the routed domains

• **Description:** Ability to create/modify **DNS records** (MX, SPF, DKIM, DMARC, TXT, CNAME) for the domains being migrated, coexisting, or verified.

• **Why required:** Domain verification, mail routing/coexistence, and authentication all depend on DNS changes.

• **How to verify:** Confirm access to the authoritative DNS zone for each domain.

• **How to implement if missing:** Coordinate with whoever controls DNS; stage changes under change control (mail-flow changes are high-impact).

4. Identity & provisioning source of truth

• **Description:** A defined **identity source of truth** (on-prem AD, Entra ID, or an HR/IGA system) and the sync mechanism (e.g. **Google Cloud Directory Sync (GCDS)** or SCIM-based provisioning).

• **Why required:** Consistent users/groups across ecosystems underpin SSO, licensing, and access.

• **How to verify:** Confirm which directory is authoritative and how objects flow into Google.

• **How to implement if missing:** Define the source of truth and provisioning model before configuring SSO or migrating.

5. A pilot scope (OU / group) and change control

• **Description:** A **pilot organizational unit or group** to validate on before fleet-wide rollout, and a **change-control** process for production changes (especially mail flow, DNS, identity, and DLP).

• **Why required:** Every solution follows an **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** model;

the pilot proves the design and change control protects production.

• **How to verify:** Confirm a named pilot OU/group exists and a change window/approval path is available.

• **How to implement if missing:** Create a pilot OU/group and agree the change-control process with the

owners before making production changes.

6. Security, compliance & governance stakeholders

• **Description:** Engagement with the **security (CISO), legal/eDiscovery, and data-governance**

stakeholders whose policies the configuration must satisfy (DLP, retention, Vault/Purview, and any regulatory regime such as ITAR/DFARS/Assured Controls where applicable).

• **Why required:** Data governance, eDiscovery, DLP, and regulated-workload settings must align to

enterprise policy and legal requirements, not just technical defaults.

• **How to verify:** Confirm the applicable policies and named stakeholders/approvers.

• **How to implement if missing:** Identify the owners and obtain the policy/compliance requirements

before configuring governance controls.

*The deployment model across every Google solution is the same: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand**, with explicit pass/fail gates and a documented rollback for production changes.*

Implementation — The Architecture-Definition Method

This is the **method** for producing and owning the target architecture, not a configuration runbook. It walks the five architecture domains — **collaboration, Drive replacement, identity, data governance & retention, and AI enablement** — and for each provides **principles**, **reference patterns**, and **architecture decision records (ADRs)**. Work the domains in dependency order: **identity first** (it is the spine — GWARCH-P2), then collaboration and Drive replacement, then governance, then AI enablement (governance must precede it — GWARCH-P6).

Each domain is authored using the same six-step loop, mapped to the **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** model:

1. **Assess** — establish current state (discovery) and constraints (NFRs, compliance, licensing).
2. **Decide** — record the domain's ADRs; every ADR states *context / decision / consequences / status* and cites the principle(s) it implements.
1. **Pattern** — capture the reference pattern(s) the workstreams must follow.
2. **Constrain** — state the non-functional requirements (NFRs) and standards the design must meet.
3. **Validate** — define the conformance gate (see `05-testing.md`).

4. **Expand** — publish to the ARB and roll to all waves.

> ***ADR numbering standard.*** `GWARCH-ADR-DOMAIN-NN` *(e.g.* `GWARCH-ADR-ID-01`*, where DOMAIN is the domain code ID/COL/DRV/GOV/AI). Every ADR has one of* `Proposed / Accepted / Superseded` *status and, if superseded, names the ADR that replaces it. This is the reusable standard the whole program cites.*

Domain 1 — Identity integration (the spine)

Design this first. Everything else resolves access through it.

Principles

• **GWARCH-ID-1 — One authoritative source, one provisioning path.** A single directory is authoritative; users and groups flow into Google through exactly one pipeline. No shadow provisioning (implements GWARCH-P2).

• **GWARCH-ID-2 — Federate authentication, automate provisioning, separate the two.** SSO (authentication) and provisioning (lifecycle) are distinct concerns designed independently so either can change without breaking the other.

• **GWARCH-ID-3 — Deprovisioning is a first-class path.** Joiner/mover/leaver flows are designed with equal rigor to onboarding; orphaned access is an architecture defect.

Reference patterns

• **Pattern ID-A — Entra ID as IdP + automated provisioning to Google.** Microsoft Entra ID federates authentication to Google Workspace via a **SAML SSO profile** (Admin console → *Security → SSO with third-party IdP → Add SAML profile*), while user/group lifecycle is provisioned via **Entra provisioning (SCIM)** or the Google connector. Legacy SSO profiles are deprecated — target SAML profiles. *(Verify current connector/SCIM behavior in your tenant.)*

• **Pattern ID-B — Directory as source of truth via GCDS.** Where on-prem AD/LDAP remains authoritative, **Google Cloud Directory Sync (GCDS)** syncs users/groups/OUs into Google on a schedule; authentication may still federate to Entra. Build the GCDS config with the Configuration Manager (it stores generated/encrypted settings) rather than hand-editing XML.

• **Pattern ID-C — Cloud Identity as the eventual authoritative directory.** For the end state where Google becomes authoritative, plan the migration of the source of authority (AD/Entra → Cloud Identity) as an explicit, sequenced ADR — not an implicit side effect.

Choose **one** primary pattern per identity population and record it; mixing patterns for the same users violates GWARCH-ID-1.

ADRs to record

• **GWARCH-ADR-ID-01** — Authoritative identity source (context: two live directories; decision: name the one source; consequences: all provisioning conforms).

• **GWARCH-ADR-ID-02** — SSO/federation model (IdP choice, SAML profile, session/MFA posture).

- **GWARCH-ADR-ID-03** — Provisioning mechanism (SCIM vs GCDS vs Admin SDK) and attribute mapping.
- **GWARCH-ADR-ID-04** — Joiner/mover/leaver + deprovisioning path and orphan-detection standard.

NFRs / standards

Provisioning latency target; SSO availability target (identity is on the critical path for 150k users); attribute-mapping standard; break-glass admin access that does **not** depend on the federated IdP (so an IdP outage cannot lock out administration).

Domain 2 — Collaboration (email, calendar, Chat, Meet, Docs)

Principles

- **GWARCH-COL-1 — Google Workspace is the single collaboration plane at end state.** Dual systems

exist only during bounded coexistence (GWARCH-P5).

- **GWARCH-COL-2 — Coexistence is routed, not replicated.** Mail/calendar coexistence uses controlled

routing and free/busy federation with a sunset date; avoid permanent bidirectional replication.

- **GWARCH-COL-3 — Collaboration policy is applied by scope.** Sharing, external chat, and Meet

settings bind to the GWARCH-Scope OU/group model (GWARCH-P3).

Reference patterns

- **Pattern COL-A — Wave-based mailbox cutover with dual-delivery bridge.** During a wave, mail routes

to both systems until MX/user cutover completes; the bridge sunsets when the wave's mailboxes are decommissioned. Calendar free/busy federates across the transition.

- **Pattern COL-B — Chat/Meet as the successor to Teams collaboration.** Define which Teams

capabilities map to Google Chat spaces vs Meet vs Drive, and which are retired, so the successor design is explicit rather than emergent.

ADRs to record

- **GWARCH-ADR-COL-01** — Coexistence routing model and per-wave sunset criteria.
- **GWARCH-ADR-COL-02** — Teams → Chat/Meet/Spaces mapping and retirement decisions.
- **GWARCH-ADR-COL-03** — External sharing/communication default posture per scope.

NFRs / standards

Mail-flow change control (high-impact); free/busy accuracy target during coexistence; a documented rollback for any MX/routing change.

Domain 3 — Content / Drive (SharePoint & OneDrive replacement)

Principles

• **GWARCH-DRV-1 — Shared drives are the unit of team content ownership.** Team content lives in

shared drives with a defined ownership/lifecycle model, not in personal My Drive, so ownership survives staff changes.

• **GWARCH-DRV-2 — Migrate by content class, not by volume.** Content is classified (active

collaboration, reference, records-retention, dark data) and each class gets a distinct migrate/archive/retire decision.

• **GWARCH-DRV-3 — Sharing is closed by default and opened by scope** (GWARCH-P4).

Reference patterns

• **Pattern DRV-A — SharePoint/OneDrive → shared-drive taxonomy mapping.** Map source sites/libraries

to a target shared-drive structure aligned to the GWARCH-Scope model; source becomes read-only, then retires, once the class is validated.

• **Pattern DRV-B — Records/retention content routed to the governance plane.** Content with retention

or hold obligations is routed to Vault-governed locations (see Domain 4), never migrated as ordinary active content.

• **Pattern DRV-C — Permission remodeling, not permission copying.** Rebuild access from the

identity/group model rather than replicating source ACLs, which propagate legacy over-sharing.

ADRs to record

• **GWARCH-ADR-DRV-01** — Shared-drive taxonomy and ownership/lifecycle standard.

• **GWARCH-ADR-DRV-02** — Content-classification scheme and per-class disposition.

• **GWARCH-ADR-DRV-03** — Permission-remodeling standard (no raw ACL copy).

NFRs / standards

Migration fidelity/validation sampling target; external-sharing default per scope; large-tenant scale considerations (150k users' content) documented as a constraint.

Domain 4 — Data governance & retention (Vault / Purview coexistence)

Design this **before** AI enablement (GWARCH-P6).

Principles

• **GWARCH-GOV-1 — Governance is a shared plane, not a per-app setting.** Retention, DLP, holds, and

classification are expressed centrally (Vault + Workspace DLP + AI control center) and inherited by every domain (implements GWARCH-P3).

• **GWARCH-GOV-2 — Legacy holds are preserved until legal sunset.** Purview holds/retention on source

data are honored until legal confirms release; the Purview↔Vault transition is an explicit, bounded coexistence (GWARCH-P5).

• **GWARCH-GOV-3 — Classify once, enforce everywhere.** A single data-classification scheme drives

DLP, sharing, retention, and AI-access decisions across all domains.

Reference patterns

• **Pattern GOV-A — Vault as the end-state retention/eDiscovery plane.** Retention rules and holds are

defined in Google Vault by scope; the retention schedule from the governance stakeholders is the input.

• **Pattern GOV-B — Bounded Purview→Vault hold transition.** Existing legal holds remain in Purview on

source data until each matter is either migrated to Vault-governed content or legally released; each hold has a sunset owner and date.

• **Pattern GOV-C — DLP by scope and classification.** Workspace DLP rules bind to the GWARCH-Scope

model and the classification scheme, closed-by-default externally.

ADRs to record

• **GWARCH-ADR-GOV-01** — End-state retention/eDiscovery plane (Vault) and schedule.

• **GWARCH-ADR-GOV-02** — Purview→Vault hold transition and sunset criteria per matter.

• **GWARCH-ADR-GOV-03** — Classification scheme and DLP enforcement standard.

NFRs / standards

Retention-schedule conformance; hold-continuity guarantee (no gap during transition); DLP false-positive/appeal path; regulated-workload controls where applicable.

Domain 5 — AI enablement (Gemini Enterprise + agentic)

Enable AI **only where** the governing controls for the data class already exist (GWARCH-P6).

Principles

• **GWARCH-AI-1 — AI inherits governance; it does not bypass it.** Gemini/agent access to Workspace

data (Gmail, Drive, Calendar, NotebookLM, Search) is constrained by the same identity, sharing, DLP, and retention controls as any user — enabled per scope through the **AI control center**.

• **GWARCH-AI-2 — Agents are governed workloads with an owned lifecycle.** Every agent has an owner,

a defined data/actions scope, and passes through the agent-platform governance surface — no ungoverned agents in production.

- **GWARCH-AI-3 — Enable by scope, measure, then expand** (GWARCH-P9). AI is piloted on a scope,

validated for governance conformance, then expanded.

Reference patterns

- **Pattern AI-A — Scoped Gemini enablement via the AI control center.** Gemini Enterprise settings

(service on/off, data-sharing) are managed in the Admin console's Generative-AI section and the AI control center; enable per OU/group in the GWARCH-Scope model rather than tenant-wide. *(Some Gemini Enterprise admin capabilities are preview/rolling-out — **verify in your tenant** and design to the capabilities actually present.)*

- **Pattern AI-B — Governed agent onboarding.** Agents that search/act on Workspace data are

registered through the agent-platform governance surface with an owner, a data-and-actions scope, and a review; access to Workspace data stores is granted per scope, not blanket.

- **Pattern AI-C — Data-access gating.** An OU/group only receives Gemini/agent access to a data class

after that class's DLP, sharing, and retention controls (Domain 4) are in place — the enforcement of GWARCH-P6.

ADRs to record

- **GWARCH-ADR-AI-01** — Gemini enablement scope model and data-sharing posture.
- **GWARCH-ADR-AI-02** — Agent governance lifecycle (registration, ownership, review, retirement).
- **GWARCH-ADR-AI-03** — AI data-access gating tied to governance readiness per class.

NFRs / standards

AI data-access auditability; agent inventory completeness; preview-capability risk register (what is GA vs preview and the contingency if a previewed capability changes).

Cross-domain: NFRs, standards, and the decision log

Beyond per-domain NFRs, the blueprint maintains program-wide standards:

- **Scale** — every design states how it holds at 150k users (limits, sync windows, admin-role

blast radius).

- **Supportability** — each domain names its operational owner and runbook location.
- **Security** — least standing access, break-glass, and closed-by-default sharing across domains.
- **Compliance** — retention/hold continuity and any regulatory regime, honored end-to-end.
- **The ADR log** — the single running record of every decision above, its status, and supersessions;

it is the artifact the ARB governs against and the one auditors read.

Once all five domains have ADRs, patterns, and NFRs recorded and validated (`05-testing.md`), compile them into the **Target-Architecture document** using the template outline in

`08-deliverables-summary.md`. That compiled document — not this method — is what the program executes against.

Testing — Architecture-Conformance Validation Gates

You do not "test" an architecture by running it; you test whether the **designs and the built estate conform** to the blueprint's principles, patterns, ADRs, and NFRs. This section defines the **conformance gates** — explicit pass/fail checks — that a domain, a wave, or the whole program must clear. `GWARCH-Validation.ps1` automates the source-side, machine-checkable subset; the rest are review gates the ARB signs.

Every gate is pass/fail with a documented remediation path. A domain does not advance from Pilot to Expand until its gates pass or the ARB records an exception ADR.

Gate 0 — Foundation conformance (blocks everything)

Check	Pass criterion
Single identity source of truth named	ADR GWARCH-ADR-ID-01 exists and is Accepted
GWARCH-Scope OU/group taxonomy defined	Named taxonomy exists incl. pilot scope
ARB + ADR log stood up	Charter and log present; cadence set
Governance stakeholders engaged	RACI + policy inputs collected

Fail → stop. No domain proceeds on an undefined foundation (GWARCH-P1/P2).

Gate 1 — Identity conformance

- **No shadow provisioning:** every population traces to the one provisioning pipeline (GWARCH-ID-1).

Check: discovery shows no workstream-owned identity source. **Pass/fail.**

- **SSO ≠ provisioning coupling:** federation and lifecycle are separable (GWARCH-ID-2).
- **Break-glass admin independent of IdP:** an admin path survives an IdP outage. **Must pass.**
- **Deprovisioning path exists and orphan detection is defined** (GWARCH-ID-3).

Gate 2 — Collaboration conformance

- **Coexistence is bounded:** every mail/calendar bridge has a sunset condition recorded

(GWARCH-COL-2 / GWARCH-P5). **Pass/fail.**

- **Collaboration policy binds to scope,** not individuals (GWARCH-COL-3).
- **MX/routing change has a documented rollback. Must pass** before any mail-flow change.

Gate 3 — Drive / content conformance

- **Team content lands in shared drives** with an ownership/lifecycle owner (GWARCH-DRV-1).

- **Content is classified and each class has a disposition** (GWARCH-DRV-2).
- **Permissions are remodeled, not raw-copied** (GWARCH-DRV-3 / no legacy over-sharing).
- **Retention-bearing content routed to the governance plane,** not migrated as active.

Gate 4 — Governance & retention conformance

- **Retention schedule expressed in Vault by scope** and matches the stakeholders' schedule (GWARCH-ADR-GOV-01).
- **Hold continuity:** no gap between Purview holds and Vault governance; each matter has a sunset owner/date (GWARCH-GOV-2). **Must pass — a hold gap is a legal defect.**
- **DLP binds to scope + classification, closed-by-default externally** (GWARCH-GOV-3 / GWARCH-P4).

Gate 5 — AI enablement conformance (gated by Gate 4)

- **AI enabled only where governance exists:** no OU/group has Gemini/agent access to a data class whose DLP/sharing/retention controls are not yet in place (GWARCH-P6 / Pattern AI-C). **Must pass.**
- **Every production agent is governed:** owner, data/actions scope, and review recorded (GWARCH-AI-2). No ungoverned agents. **Pass/fail on the agent inventory.**
- **Preview-capability risk register maintained:** each Gemini capability marked GA vs preview with a contingency (**verify in tenant** items are tracked, not assumed GA).
- **AI enablement is scoped, not tenant-wide,** and auditable (GWARCH-AI-1/AI-3).

Gate 6 — Cross-domain NFR conformance

- **Scale stated at 150k users** per domain (limits, sync windows, admin blast radius).
- **Every domain names an operational owner + runbook** (supportability).
- **Least standing access + break-glass** verified across domains (security).
- **Compliance continuity** end-to-end (retention/holds/regulatory).

Gate 7 — ADR completeness & authority

- **Every domain has its required ADRs recorded** with status; supersessions are linked.
- **No un-reconciled design conflicts:** any workstream deviation from a principle/pattern is either removed or carried as an explicit exception ADR (GWARCH-P1). **Pass/fail.**
- **The ADR log is the single source** the ARB governs against and is current.

How to run the machine-checkable subset

`GWARCH-Validation.ps1` performs read-only source-side checks that feed Gates 0–2 (e.g. detecting multiple potential identity sources, unscoped targeting, missing break-glass indicators) and emits a pass/fail summary. `GWARCH-Report.ps1` rolls the results plus the design-review gates into a **coverage/gap** view the ARB uses to decide Pilot→Expand. The review gates (patterns, ADRs, NFR sign-off) are human ARB decisions the scripts surface but do not replace.

Exit criterion for "architecture approved": Gates 0–7 pass (or carry ARB-approved exceptions), the Target-Architecture document is compiled, and the ADR log is current. Only then does the architecture Expand to all waves.

Troubleshooting — Common Architecture Pitfalls

These are the failure modes that erode an owned target architecture in a large Google Workspace + Gemini transformation, with the symptom, root cause, and the corrective move. Each references the principle or gate it protects.

1. Competing identity sources of truth (shadow provisioning)

- **Symptom:** duplicate/mismatched users, licensing gaps, access that "shouldn't exist." Discovery finds more than one thing creating identities.
- **Root cause:** a workstream provisioned its own users instead of consuming the one pipeline — violates **GWARCH-ID-1**.
- **Fix:** name the single authoritative source (ADR GWARCH-ADR-ID-01), collapse shadow sources onto the one provisioning path, and re-run `GWARCH-Validation.ps1` until Gate 1 passes. Prevent recurrence with the identity contract in `03`.

2. Break-glass admin depends on the federated IdP

- **Symptom:** a plausible scenario where an Entra/IdP outage locks administrators out of a 150k-user Google estate.
- **Root cause:** SSO and administrative access were coupled — violates **GWARCH-ID-2** and the Gate 1 "must pass" break-glass check.
- **Fix:** stand up an emergency super-admin path that does **not** traverse the federated IdP; test it after every federation change. Never ship a federation change without this.

3. Coexistence bridges that never sunset

- **Symptom:** mail dual-delivery, directory bridges, or legacy holds still running long after the wave they supported completed.
- **Root cause:** the bridge had no sunset condition — violates **GWARCH-P5 / GWARCH-COL-2**.
- **Fix:** attach a written sunset condition and owner to every bridge; review the list at each ARB;

escalate any bridge past its date. Gate 2 fails until every bridge is bounded.

4. Permissions copied instead of remodeled (legacy over-sharing)

• **Symptom:** post-migration, content is over-shared exactly as it was in SharePoint/OneDrive; external exposure appears in the target.

• **Root cause:** source ACLs were replicated rather than rebuilt from the group model — violates **GWARCH-DRV-3** and closed-by-default (**GWARCH-P4**).

• **Fix:** remodel access from the identity/group taxonomy (Pattern DRV-C), default external sharing closed per scope, and validate a sample at Gate 3.

5. AI enabled ahead of governance

• **Symptom:** Gemini/agents can reach a data class whose DLP, sharing, or retention controls aren't in place yet.

• **Root cause:** AI enablement outran governance — violates **GWARCH-P6 / Pattern AI-C**; fails the Gate 5 "must pass" check.

• **Fix:** disable Gemini/agent access for that class until Domain 4 controls exist; enable per scope through the AI control center only after the data-class governance is validated. Track the gate in the ADR log.

6. Ungoverned agents in production

• **Symptom:** agents acting on Workspace data with no recorded owner, scope, or review.

• **Root cause:** no agent lifecycle — violates **GWARCH-AI-2**.

• **Fix:** register every production agent through the agent-platform governance surface with an owner and a data/actions scope; retire orphans. The agent inventory must be complete for Gate 5.

7. Designing on preview capabilities as if GA

• **Symptom:** a design depends on a Gemini Enterprise feature that changes, disappears, or behaves differently than the screenshot it was drawn from.

• **Root cause:** a preview/rolling-out capability was assumed GA instead of **verified in tenant**.

• **Fix:** maintain the preview-capability risk register (GA vs preview + contingency); design to the capabilities actually present in your tenant; re-verify at each ARB. Never invent behavior.

8. Policy applied to individuals, not scopes

• **Symptom:** inconsistent DLP/sharing/retention/Gemini access; exceptions multiply; new capabilities land ungoverned.

- **Root cause:** governance by exception instead of by scope — violates **GWARCH-P3**.
- **Fix:** refactor ad-hoc, individual-level policy onto the GWARCH-Scope OU/group model; keep source→target scope mapping deterministic (`GWARCH-Set-ScopeTargeting.ps1`). Treat any individual-level policy as a defect.

9. Decisions made outside the ADR log

- **Symptom:** two workstreams build conflicting things; no one can say why a choice was made; auditors find undocumented design.
- **Root cause:** decisions live in slides/email, not the ADR log — erodes **GWARCH-P1** authority.
- **Fix:** every consequential decision becomes an ADR (supersede, never overwrite); the ARB governs against the single log; Gate 7 fails on un-reconciled conflicts.

10. The architecture drifts from the real estate

- **Symptom:** the compiled document, the ADR log, and the built estate disagree.
- **Root cause:** no re-baselining; the design was validated once and never re-measured (**GWARCH-P8**).
- **Fix:** re-run `GWARCH-Discovery.ps1` + `GWARCH-Validation.ps1` on a cadence; feed drift back as gaps into the ADR log and ARB; keep the three surfaces coherent (`07` §9).

Script-level issues

- `Connect-MgGraph` **fails / insufficient scopes:** consent the exact read scopes listed in each script header; a Global Reader role covers the discovery scopes. Re-run after consent.
- **Empty or partial output at 150k scale:** discovery/report scripts page and can run long on a large tenant; let them complete, run from a stable session, and export to file rather than console.
- **Remediation script "did nothing":** by design — `GWARCH-Remediation.ps1` defaults to `-WhatIf` and only prints guidance; it makes no changes unless explicitly run with `-WhatIf:$false` after review.

Best Practices — Keeping the Architecture Authoritative

Defining the target architecture is the easy half; keeping it **authoritative** across a multi-year, multi-partner, 150k-user program is the hard half. These practices keep the blueprint the thing every workstream conforms to instead of a document that drifts into irrelevance.

1. Make the ADR the currency of decisions

- **Every consequential decision becomes an ADR** (`GWARCH-ADR-DOMAIN-NN`, status

Proposed/Accepted/Superseded). If a decision isn't in the log, it isn't a decision — it's a rumor.

• **Never edit a decision in place; supersede it.** A new ADR references the one it replaces. The history is the audit trail.

• **The ADR log is singular and current.** One log, owned by the ARB, is the authority — not slides, not email threads.

2. Govern by scope, everywhere (GWARCH-Scope discipline)

• Bind **every** policy — sharing, DLP, retention, Gemini access — to the named OU/group taxonomy, so new capabilities inherit governance by landing in the right scope.

• Treat a policy applied to an ad-hoc individual as a **defect**; refactor it onto a scope.

• Keep source-side group→target-scope mapping deterministic (`GWARCH-Set-ScopeTargeting.ps1`) so scope can't silently diverge between the M365 source and the Google target.

3. Identity-first, always

• Resolve identity ADRs before any dependent domain proceeds (GWARCH-P2). A late identity change invalidates downstream designs across four domains.

• Keep **break-glass admin access independent of the federated IdP** — verify it after every federation change. An IdP outage must never lock out administration of a 150k-user estate.

4. Keep coexistence on a leash

• Give **every** coexistence bridge (mail routing, directory, holds) a written sunset condition and an owner (GWARCH-P5). Review the list at every ARB.

• A "temporary" bridge past its sunset date is an escalation item, not a footnote.

5. Governance precedes AI — hold the line

• Do not let AI enablement outrun governance (GWARCH-P6). The Pattern AI-C gate is non-negotiable: no Gemini/agent access to a data class until that class's DLP/sharing/retention controls exist.

• Maintain a **preview-capability risk register.** Gemini Enterprise capabilities move fast and some are preview/rolling-out — mark each GA vs preview, carry a contingency, and **verify in tenant** rather than designing on a screenshot.

• Keep a complete **agent inventory** with owners; an ungoverned production agent is a governance breach, not a convenience.

6. Run the ARB as a conformance engine

- **Standing cadence, quorum, decision rights.** The ARB approves ADRs, grants exceptions (as

exception ADRs), and owns the sunset list.

- **ARB hooks into the program:** wave gates (`05-testing.md`) feed the ARB; a wave doesn't advance

Pilot→Expand without an ARB conformance sign-off. This is the enforcement mechanism that makes the blueprint authoritative rather than advisory.

- **Exceptions are visible and time-boxed.** Every deviation is an explicit, dated exception ADR with

a remediation plan — never a silent divergence.

7. Re-baseline against reality on a cadence

- Re-run `GWARCH-Discovery.ps1` and `GWARCH-Validation.ps1` on a schedule so the architecture is

measured against the *current* source estate, not the one it was drafted against (GWARCH-P8).

- Feed drift (new shadow identity sources, unscoped policies, expired coexistence bridges) back into

the ADR log and the ARB as gaps to close.

8. Design for reversibility where it's cheap

- Prefer exportable formats and portable identity patterns (GWARCH-P7) so a future tooling change

touches integration layers, not the whole estate. Record any deliberate one-way lock-in as an ADR with its rationale so it's a chosen risk, not an accident.

9. Keep the three surfaces coherent

The **compiled Target-Architecture document**, the **ADR log**, and the **built estate** must agree. When the estate changes, the ADR log and document change with it (supersede, don't overwrite). A gap between the three is exactly what erodes authority — close it before it compounds.

Bottom line: authority comes from **one owned end-state, one ADR log, scope-based governance, an ARB that gates every wave, and periodic re-baselining against the real estate.* Hold those and the architecture stays the thing the program conforms to — which is the entire point of owning it.

Deliverables Summary

This blueprint's purpose is to produce **one authoritative artifact** — the compiled **Target-Architecture document** — governed by an **ADR log** and enforced by an **ARB**. This section lists what the solution delivers and provides the **Target-Architecture document template outline** you populate to create that artifact.

What the solution delivers

Deliverable	Purpose
Solution overview (`01`)	The end-state architecture, how workstreams fit, and the enterprise principles (GWARCH-P1…P9)
Requirements (`02`)	Prerequisites to run the architecture-authority function (tenants, identity source of truth, scope taxonomy, governance stakeholders, ARB, tooling)
Architecture-definition method (`03`)	The per-domain method — principles, reference patterns, and ADRs for identity, collaboration, Drive replacement, governance/retention, AI enablement
Conformance gates (`05`)	Pass/fail validation gates (0–7) a domain/wave/program must clear
Troubleshooting (`06`)	Common architecture pitfalls and how to correct them
Best practices (`07`)	Keeping the architecture authoritative; ARB hooks
This summary + template (`08`)	Deliverables list and the Target-Architecture document template outline
5 source-side scripts	Discovery, conformance validation, coverage/gap report, remediation guidance, scope targeting (Microsoft Graph, read-only where they read live systems)
Executive summary	One-page framing for sponsors/ARB

The five architecture domains (recap)

Identity (the spine) · Collaboration · Drive / content replacement · Data governance & retention · AI enablement. Each carries principles, reference patterns, ADRs, and NFRs (see `03`), and is gated for conformance (see `05`).

Target-Architecture Document — template outline

Populate this outline to produce the authoritative document the program executes against. Keep it white-label and reusable; every section cites the principles/ADRs it implements.

0. Front matter

- Document owner (chief/enterprise architect), version, status, ARB approval date, supersession note.

1. Executive summary

- The end state in one page: Google Workspace + Gemini + one identity spine; why it's owned centrally;

the bottom-line outcome (every workstream aligned to one end-state design so waves execute without architectural rework).

2. Context, scope & drivers

- Transformation scope (150k users, M365 → Google Workspace + Gemini), business drivers, in/out of

scope, key constraints and assumptions.

3. Architecture principles

- The enterprise principles (GWARCH-P1…P9) verbatim, each with its rationale and implications. These

constrain everything that follows.

4. Current-state summary

- Source M365/Entra estate as discovered (`GWARCH-Discovery.ps1` output): identity sources, mail,

content, governance/holds, tenant/licensing. The reality the design starts from.

5. Target-state architecture — per domain For each of the five domains (Identity, Collaboration, Drive replacement, Governance/Retention, AI enablement):

- 5.x.1 **Principles** (domain principle set)
- 5.x.2 **Reference pattern(s)** chosen, with a diagram
- 5.x.3 **Key ADRs** (`GWARCH-ADR-DOMAIN-NN`, with status)
- 5.x.4 **Non-functional requirements & standards**
- 5.x.5 **Coexistence & sunset** (where applicable)

6. Integration contracts

- The identity contract (one provisioning path), the OU/group targeting contract (GWARCH-Scope), and

the governance contract (shared plane) that bind the domains together.

7. Non-functional requirements (cross-domain)

- Scale (150k), availability, supportability, security (least standing access, break-glass),

compliance (retention/hold continuity, regulatory).

8. Governance & the ARB

- ARB charter, cadence, decision rights; how wave gates feed the ARB; exception-ADR process.

9. ADR log

- The complete running decision record (or a pointer to it), every ADR with status and supersessions.

10. Conformance & validation

- The gates (0–7) and how designs/waves are measured; how the scripts feed them.

11. Roadmap & wave alignment

- How the architecture maps onto the wave plan (Assess → Configure → Pilot → Validate → Expand) and

what each wave must satisfy.

12. Risk register

- Including the **preview-capability register** (Gemini GA vs preview / "verify in tenant" items with

contingencies) and any deliberate lock-in decisions.

13. Appendices

- Glossary of named standards (GWARCH-Scope, ADR numbering, principle IDs), stakeholder RACI, and the

script outputs referenced.

Definition of done: the Target-Architecture document is compiled from the domain work in `03`, all conformance gates in `05` pass or carry ARB exceptions, the ADR log is current, and the ARB has approved it. At that point the architecture is authoritative and Expands to all waves.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

GWARCH-Discovery.ps1

Purpose: Inventories the source Microsoft 365 / Entra ID estate that the Google Workspace + Gemini target architecture must be designed against: identity sources and provisioning signals, domains and mail routing, groups/OU-equivalent structure, licensing, and governance/hold indicators. Output grounds the Assess phase so the architecture is drawn from reality. Read-only: it never changes the source tenant. Provide your-tenant-id at the prompt or via -TenantId. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\GWARCH-Discovery.ps1 powershell -ExecutionPolicy Bypass -File .\GWARCH-Discovery.ps1 -TenantId your-tenant-id -OutputPath .\gwarch-discovery.json Requires: Microsoft.Graph PowerShell SDK. Graph scopes (read-only): Directory.Read.All, User.Read.All, Group.Read.All, Organization.Read.All, Domain.Read.All, Policy.Read.All. EXPECTED RESULTS A console summary plus a JSON report of the current-state source estate (counts, domains, licensing, provisioning/hold indicators). No pass/fail — this is factual input to the design. .NOTES White-label — the repeatable way. Read-only where it reads live systems. #>

How to run: powershell -ExecutionPolicy Bypass -File .\GWARCH-Discovery.ps1 powershell -ExecutionPolicy Bypass -File .\GWARCH-Discovery.ps1 -TenantId your-tenant-id -OutputPath .\gwarch-discovery.json Requires: Microsoft.Graph PowerShell SDK. Graph scopes (read-only): Directory.Read.All, User.Read.All, Group.Read.All, Organization.Read.All, Domain.Read.All, Policy.Read.All. EXPECTED RESULTS A console summary plus a JSON report of the current-state source estate (counts, domains, licensing, provisioning/hold indicators). No pass/fail — this is factual input to the design. .NOTES White-label — the repeatable way. Read-only where it reads live systems. #>

GWARCH-Remediation.ps1

Purpose: Turns the FAIL findings from GWARCH-Validation.ps1 into ordered, safe remediation guidance for closing target-architecture conformance gaps (single identity source, break-glass admin, scope-based targeting, bounded coexistence). It defaults to -WhatIf and PRINTS the recommended steps and any candidate source-side action without changing anything. Most fixes are design or target-side decisions and are deliberately surfaced as guidance for ARB action, never auto-applied. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\GWARCH-Remediation.ps1 powershell -ExecutionPolicy Bypass -File .\GWARCH-Remediation.ps1 -ValidationPath .\gwarch-validation.json By default nothing is changed (-WhatIf on). To allow any explicitly supported source-side action, re-run with -WhatIf:$false AFTER ARB review. Graph scopes only needed if you opt into an action. EXPECTED RESULTS A prioritized remediation plan per FAIL. In default -WhatIf mode: guidance only, no changes. pass = every referenced gap has an owner and a recorded next step (ADR or fix). .NOTES White-label — the repeatable way. Read-only where it reads live systems. #>

How to run: powershell -ExecutionPolicy Bypass -File .\GWARCH-Remediation.ps1 powershell -ExecutionPolicy Bypass -File .\GWARCH-Remediation.ps1 -ValidationPath .\gwarch-validation.json By default nothing is changed (-WhatIf on). To allow any explicitly supported source-side action, re-run with -WhatIf:$false AFTER ARB review. Graph scopes only needed if you opt into an action. EXPECTED RESULTS A prioritized remediation plan per FAIL. In default -WhatIf mode: guidance only, no changes. pass = every referenced gap has an owner and a recorded next step (ADR or fix). .NOTES White-label — the repeatable way. Read-only where it reads live systems. #>

GWARCH-Report.ps1

Purpose: Combines the current-state discovery and the conformance-validation outputs into a single coverage/gap report across the five architecture domains (identity, collaboration, Drive replacement, data governance/retention, AI enablement). Shows per-domain readiness and the open gaps the Architecture Review Board weighs for the Pilot-to-Expand decision. Read-only: it reads the JSON reports the other scripts produced; it does not touch live systems. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\GWARCH-Report.ps1 powershell -ExecutionPolicy Bypass -File .\GWARCH-Report.ps1 -DiscoveryPath .\gwarch-discovery.json -ValidationPath .\gwarch-validation.json -OutputPath .\gwarch-coverage-report.md Requires: the JSON outputs of GWARCH-Discovery.ps1 and GWARCH-Validation.ps1 (no Graph scopes). EXPECTED RESULTS A Markdown coverage/gap report (per-domain readiness + open gaps + FAIL count) the ARB uses to gate a wave. No live-system changes. .NOTES White-label — the repeatable way. Read-only where it reads live systems. #>

How to run: powershell -ExecutionPolicy Bypass -File .\GWARCH-Report.ps1 powershell -ExecutionPolicy Bypass -File .\GWARCH-Report.ps1 -DiscoveryPath .\gwarch-discovery.json -ValidationPath .\gwarch-validation.json -OutputPath .\gwarch-coverage-report.md Requires: the JSON outputs of GWARCH-Discovery.ps1 and GWARCH-Validation.ps1 (no Graph scopes). EXPECTED RESULTS A Markdown coverage/gap report (per-domain readiness + open gaps + FAIL count) the ARB uses to gate a wave. No live-system changes. .NOTES White-label — the repeatable way. Read-only where it reads live systems. #>

GWARCH-Set-ScopeTargeting.ps1

Purpose: Builds and validates the deterministic mapping from source Microsoft Entra groups to the target GWARCH-Scope OU/group taxonomy (pilot, per-wave, regulated scopes) that every policy binds to (GWARCH-P3). By default it PREVIEWS the mapping (-WhatIf) and writes a plan file; it makes no change to the source. It can optionally tag matched source groups' membership export for handoff to the provisioning pipeline only when explicitly run with -WhatIf:$false. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\GWARCH-Set-ScopeTargeting.ps1 -MappingPath .\scope-map.csv powershell -ExecutionPolicy Bypass -File .\GWARCH-Set-ScopeTargeting.ps1 -MappingPath .\scope-map.csv -WhatIf:$false MappingPath CSV columns: SourceGroup,TargetScope,ScopeType. Requires Microsoft.Graph SDK. Graph scopes (read-only preview): Group.Read.All, GroupMember.Read.All, Directory.Read.All. EXPECTED RESULTS A resolved scope-targeting plan (source group -> target scope) with match/miss counts, written to a plan file. In default -WhatIf mode: preview only, no change. pass = every intended scope has a matched source group and no source group is unmapped/ambiguous. .NOTES White-label — the repeatable way. Read-only where it reads live systems. #>

How to run: powershell -ExecutionPolicy Bypass -File .\GWARCH-Set-ScopeTargeting.ps1 -MappingPath .\scope-map.csv powershell -ExecutionPolicy Bypass -File .\GWARCH-Set-ScopeTargeting.ps1 -MappingPath .\scope-map.csv -WhatIf:$false MappingPath CSV columns: SourceGroup,TargetScope,ScopeType. Requires Microsoft.Graph SDK. Graph scopes (read-only preview): Group.Read.All, GroupMember.Read.All, Directory.Read.All. EXPECTED RESULTS A resolved scope-targeting plan (source group -> target scope) with match/miss counts, written to a plan file. In default -WhatIf mode: preview only, no change. pass = every intended scope has a matched source group and no source group is unmapped/ambiguous. .NOTES White-label — the repeatable way. Read-only where it reads live systems. #>

GWARCH-Validation.ps1

Purpose: Runs the machine-checkable subset of the target-architecture conformance gates against the source Microsoft 365 / Entra ID estate: single identity source of truth (no shadow provisioning), break-glass admin independent of the federated IdP, scope-based targeting, and bounded-coexistence indicators. Emits pass/fail per check to feed Gates 0-2 of the blueprint. Read-only: it evaluates, it never changes the source. Provide your-tenant-id via -TenantId. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\GWARCH-Validation.ps1 powershell -ExecutionPolicy Bypass -File .\GWARCH-Validation.ps1 -TenantId your-tenant-id -OutputPath .\gwarch-validation.json Requires: Microsoft.Graph SDK. Graph scopes (read-only): Directory.Read.All, User.Read.All, Group.Read.All, Organization.Read.All, RoleManagement.Read.Directory, Policy.Read.All. EXPECTED RESULTS A pass/fail table per conformance check plus a JSON report. PASS = conformant; FAIL = a gap to remediate (see GWARCH-Remediation.ps1) or record as an ARB exception ADR. .NOTES White-label — the repeatable way. Read-only where it reads live systems. #>

How to run: powershell -ExecutionPolicy Bypass -File .\GWARCH-Validation.ps1 powershell -ExecutionPolicy Bypass -File .\GWARCH-Validation.ps1 -TenantId your-tenant-id -OutputPath .\gwarch-validation.json Requires: Microsoft.Graph SDK. Graph scopes (read-only): Directory.Read.All, User.Read.All, Group.Read.All, Organization.Read.All, RoleManagement.Read.Directory, Policy.Read.All. EXPECTED RESULTS A pass/fail table per conformance check plus a JSON report. PASS = conformant; FAIL = a gap to remediate (see GWARCH-Remediation.ps1) or record as an ARB exception ADR. .NOTES White-label — the repeatable way. Read-only where it reads live systems. #>